

Subject Description Form

Subject Code	EEE511
Subject Title	Fundamental Quantum Information
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	The objective of this subject is to provide students with a comprehensive understanding of the principles and applications of quantum information science. This subject aims to introduce students to the fundamental concepts of quantum information, quantum computation, and quantum communication. Students will learn about quantum states, quantum gates, and quantum algorithms, exploring how these concepts differ from classical information processing.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understanding the principles of quantum information including the mathematical description. - Analyzing quantum computational models such as Shor and Grover algorithm. - Applying quantum information principles to problem-solving and performing calculations related to quantum states, quantum gates, and quantum algorithms. - Explaining quantum communication protocols. - Be able to evaluate the potential impact of quantum information technologies in various fields, such as cryptography, simulation, and optimization.
Subject Synopsis/ Indicative Syllabus	Introduction to quantum information: Review of key concepts and mathematical formalism of quantum mechanics. Quantum states and measurements: Understanding quantum states, superposition, entanglement, and measurement theory. Quantum gates and circuits: Study of quantum gates, unitary transformations, and quantum circuit representation. Quantum information processing: Introduction to quantum algorithms, including quantum Fourier transform, Grover's algorithm, and Shor's algorithm. Quantum communication: Principles of quantum communication, quantum teleportation, quantum cryptography, and quantum key distribution. Quantum error correction: Basics of quantum error correction codes and fault-tolerant quantum computing.

Teaching/Learning Methodology	Lecture: The recent advancements in quantum information experiments will be explained, teaching the principles and calculation steps to understand the fundamentals of quantum information combined with exercise. Assignments will be given to assess students' learning progress. Tutorial: A selection of problems and group discussion topics will be provided. Students are strongly encouraged to attempt solving the problems prior to receiving the solutions.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Continuous Assessments (Assignments and tests)</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Examination</td><td>60%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Continuous assessments consist of assignments and a mid-term test. These will assess the students’ understanding of basic concepts and principles in quantum information. The examination covers the main contents in this subject, including the essential knowledge of quantum information experiments, quantum error correction codes and fault-tolerant quantum computing.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Continuous Assessments (Assignments and tests)	40%	√	√	√	√	√	2. Examination	60%	√	√	√	√	√	Total	100 %					
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Reading List and References	Textbook • Nielsen, Michael A., and Isaac L. Chuang. Quantum computation and quantum information. Cambridge university press, 2010. References • Petz, Dénes. Quantum information theory and quantum statistics. Springer Science & Business Media, 2007. • Yanofsky, Noson S., and Mirco A. Mannucci. Quantum computing for computer scientists. Cambridge University Press, 2008. • Hayashi, Masahito. Quantum information theory. New York: Springer, 2016.
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